

PROJECT REPORT

CyberShield

A Multi-Modal Phishing Detection Platform

URL Analysis, SMS/Email NLP Analysis, and Screenshot Phishing Detection

Submitted in partial fulfillment of the academic project requirements

Version 5.0

Acknowledgement

The CyberShield team sincerely thanks its project guide, faculty members, classmates, families, and friends for their support during the development of this academic project. Their guidance helped shape the project from a URL-focused detector into a broader phishing-awareness platform that considers the way people actually encounter suspicious content: through links, messages, emails, and screenshots.

The team also acknowledges the cybersecurity and research communities whose standards, threat reports, secure-development guidance, and open technical work make responsible phishing analysis possible. The report is prepared using the college-provided project-report index so that its study, design, data, results, and references can be reviewed in a familiar academic structure.

Abstract

CyberShield is a web-based, multi-modal phishing detection platform designed to help users assess suspicious URLs, SMS messages, email content, and screenshots before they interact with them. The system combines cybersecurity intelligence, natural-language processing (NLP), and image processing (IP) to identify technical, linguistic, and visual indicators of phishing. Unlike a link-only checker, CyberShield produces a unified risk assessment with evidence, confidence, limitations, and safe next steps.

The proposed platform accepts a user submission, validates and normalizes it, applies a modality-specific analysis pipeline, combines the resulting signals, and displays an understandable result. It is designed around privacy-aware storage, role-based access, audit logging, safe file handling, configurable retention, and measurable verification. This report documents the problem, technology choices, system requirements, diagrams, data design, results, limitations, and future scope of CyberShield.

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1 Introduction

1.1 Introduction

Phishing is no longer limited to a plainly suspicious web address. Attackers use shortened or lookalike URLs, urgent SMS messages, polished email text, copied logos, QR codes, and screenshots that imitate legitimate login pages. Users often have only seconds to decide whether to click, reply, scan, or share a one-time password. CyberShield is proposed as a single platform that analyzes the evidence available in each of these forms and gives a cautious, understandable risk assessment before an unsafe action is taken.

CyberShield treats this capability as a controlled pipeline rather than a single prediction. Input is validated, normalized, and routed only to the services required for the selected modality. Each analyzer returns versioned indicators with severity, evidence source, and confidence. The scoring layer removes duplicate evidence, applies a transparent policy, and records the limitations of unavailable services. This separation makes the behavior easier to test, protects sensitive content, and allows a future model or intelligence-feed change to be reviewed without changing the entire platform.

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1 Introduction

1.2 Aims and Objectives of the Work

The aim of CyberShield is to create a reliable and user-oriented phishing detection platform that supports URL analysis, SMS/email text analysis, and screenshot analysis. Its objectives are to identify commonly observed phishing indicators, combine cybersecurity and AI-based evidence, explain the reason for a risk level, preserve privacy, support feedback, and provide a foundation for future integration with security teams. The project emphasizes prevention and awareness rather than automated enforcement or unverified accusation.

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1 Introduction

1.3 Brief Literature Review

Phishing research and operational guidance consistently show that social engineering succeeds by creating urgency, trust, fear, or reward. URL-based signals such as deceptive domains and malicious reputation remain important, but language analysis adds evidence about credential requests and impersonation. OCR and visual analysis extend coverage to screenshots that conceal a destination or reproduce a trusted brand. This review motivates a multi-modal design in which no single weak signal is treated as proof, while several corroborating signals can produce a strong warning.

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1 Introduction

1.4 Problem Definition

Existing link checkers do not fully address the way suspicious content is shared through messages and images. A user may receive a delivery SMS with a link, a fake banking email without a visible URL, or a screenshot asking for an OTP. In each case, the user needs a prompt assessment that identifies the relevant evidence and suggests a safe response. The central problem is therefore to provide a privacy-conscious, explainable analysis workflow that can interpret multiple input forms without automatically visiting dangerous content or promising perfect detection.

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1 Introduction

1.5 Plan of the Work

The project plan begins with requirement study and threat understanding, followed by technology selection, architecture design, dataset preparation, implementation, testing, and documentation. URL analysis provides the initial cybersecurity layer; the NLP module extends detection to SMS and email language; the image module adds OCR and visual indicators for screenshots. Integration work combines results through a versioned scoring policy. The final phase evaluates accuracy, usability, performance, security, and limitations before preparing the report and demonstration.

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2 Technology and Literature Review

2.1 Cybersecurity Foundations

CyberShield uses defensive cybersecurity principles: untrusted input is validated, suspicious URLs are never opened in the user browser, reputation lookups are server-side, and results are presented as decision support. Important technical indicators include host structure, top-level domain, encoding, redirect behavior, certificate-related context, and threat-intelligence reputation.

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2 Technology and Literature Review

2.2 URL Analysis Technologies

A URL parser separates scheme, host, path, query, port, and encoded characters. Normalization detects misleading forms such as IP-address hosts, repeated separators, punycode, long subdomains, or lookalike brand names. Threat-intelligence adapters enrich the normalized indicator while retaining source freshness and failure status.

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2 Technology and Literature Review

2.3 NLP for SMS and Email

NLP is used to detect urgency, threats, prize claims, account-verification requests, credential or payment requests, sender impersonation, and unusual instructions. Entity extraction identifies domains, phone numbers, email addresses, amounts, and organization names. The approach combines rules and cautious classification so that ordinary urgent communication is not automatically called fraud.

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2 Technology and Literature Review

2.4 Image Processing and OCR

Screenshot analysis begins with safe decoding of PNG, JPEG, or WEBP files. OCR extracts visible text and confidence values; image processing identifies form-like layouts, QR codes, login prompts, payment prompts, and logo-like patterns. Extracted text is passed to the NLP service, while visual signals remain distinct in the explanation.

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2 Technology and Literature Review

2.5 Web Application Technologies

The platform can use a responsive web client, a secure API layer, asynchronous analysis workers, a relational database, protected object storage, and monitoring. Containerized services allow URL, NLP, and image workloads to scale independently. HTTPS, server-side secrets, structured logging, and role-based authorization are baseline engineering requirements.

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2 Technology and Literature Review

2.6 Literature Review Findings

The literature and professional guidance support a defense-in-depth strategy. Technical indicators identify known malicious infrastructure; linguistic signals identify social-engineering pressure; visual analysis identifies copied interfaces or hidden text. The gap addressed by CyberShield is a user-facing academic platform that brings these evidence types together and makes uncertainty visible rather than hiding it behind a single unexplained label.

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2 Technology and Literature Review

2.7 Technology Selection Rationale

Technology choices prioritize explainability, safe integration, and maintainability. Deterministic rules and parsers provide a reviewable baseline; external intelligence is optional enrichment; models are versioned and evaluated before use. The system is designed so that a degraded provider lowers confidence instead of making unsafe content appear clean.

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3 System Requirements Study

3.1 User Characteristics

CyberShield serves different user groups with different information needs. A guest can submit a limited analysis and see the immediate result. A registered user can manage history and consent. An analyst reviews authorized flagged cases and provides disposition. An administrator configures limits, policies, roles, and integrations. An auditor reviews permitted governance events. Role separation reduces exposure of raw content and ensures that users see only the information required for their tasks.

Table 1: User Characteristics

User class	Primary activity	Access boundary
Guest	Analyze one submitted item	No retained history by default.
Registered user	Analyze and view own history	Own records and privacy settings only.
Analyst	Review authorized cases	Assigned or policy-approved cases.
Administrator	Manage policy and operations	Configuration and audited sensitive access.
Auditor	Review governance records	Read-only, permitted audit scope.

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3 System Requirements Study

3.2 Hardware Requirements

The academic deployment requires a modern client device with an updated browser and stable internet connection. The server environment requires sufficient CPU and memory for concurrent requests, protected storage for optional screenshots, and optional GPU capacity only when a selected image model needs it. Resource limits prevent large or deliberately malformed uploads from consuming analysis capacity.

Table: Supporting Requirement Baseline

Item	Requirement
Component	Baseline requirement
Client	Desktop/mobile browser, 4 GB RAM, internet connection
Application server	2 CPU cores, 4 GB RAM minimum
Worker node	4 CPU cores, 8 GB RAM for OCR/NLP workload
Storage	Encrypted database backup and protected object storage

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3 System Requirements Study

3.3 Software Requirements

The platform requires a supported operating system, modern browser, HTTPS-capable web server, relational database, background queue, OCR engine, and monitored logging stack. Development uses a version-control system, test framework, dependency scanner, and secure configuration method. All production secrets remain outside source code and local test files.

Table: Supporting Requirement Baseline

Item	Requirement
Layer	Suggested responsibility
Frontend	Responsive browser interface and accessibility
Backend	Authentication, API, scoring, audit
NLP service	Text normalization and phishing cues
Image service	Safe image decode, OCR, visual signals
Database	Users, requests, results, indicators, audit

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3 System Requirements Study

3.4 Functional Requirements: URL Analysis

The system shall accept a URL as text, validate its syntax, normalize meaningful components, identify lexical and structural indicators, and query configured reputation sources without opening the target in the user browser. The result shall include risk level, score, confidence, reasons, source limitations, and safe next actions. Redirect evaluation is optional and may run only inside an approved sandbox with strict network and time limits.

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3 System Requirements Study

3.5 Functional Requirements: SMS and Email Text

The system shall accept pasted SMS text, email subject/body content, and available sender fields. It shall detect credential or payment requests, urgency, threats, impersonation claims, suspicious embedded entities, and language limitations. Sensitive values are masked in displayed evidence and are not written to logs. The result shall explain which phrases or extracted links influenced the assessment.

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3 System Requirements Study

3.6 Functional Requirements: Screenshot Analysis

The system shall accept approved image formats within configured limits, verify the file signature, decode pixels safely, perform OCR, identify relevant visual phishing cues, and state OCR confidence. It shall flag login and OTP prompts, payment prompts, QR codes, obscured links, brand-like patterns, and suspicious urgency banners. Image-derived and OCR-derived evidence are displayed separately.

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3 System Requirements Study

3.7 Functional Requirements: Accounts and Feedback

Registered users shall sign in through the approved identity mechanism, view only their own history, manage retention and consent preferences, delete eligible records, and submit feedback. Feedback distinguishes unclear explanation, possible false positive, and possible false negative. Analysts may add a documented disposition without changing the immutable original score record.

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3 System Requirements Study

3.8 Functional Requirements: Administration

Administrators shall manage user roles, enabled integrations, input limits, retention rules, risk thresholds, and rule/model versions. Administrative changes, sensitive asset access, exports, deletion actions, and analyst overrides shall create immutable audit events. The administrative dashboard shall distinguish ordinary workload, external-feed degradation, queued work, service failure, and suspicious access activity.

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3 System Requirements Study

3.9 Non-functional Requirements: Performance

A typical URL or text result should be returned within five seconds at the defined 95th-percentile baseline. A supported screenshot should return within fifteen seconds, excluding a disclosed external-source timeout. The system shall queue long-running work rather than leaving the browser connection indefinitely open, and it shall show progress status without exposing implementation secrets.

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3 System Requirements Study

3.10 Non-functional Requirements: Security

The system shall enforce TLS, least-privilege role-based authorization, secure session management, input validation, rate limiting, protected object storage, server-side secrets, and safe logging. It shall defend against common application risks such as injection, cross-site scripting, broken access control, insecure upload, and accidental disclosure of credentials. Failed dependencies must not silently produce a safe label.

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3 System Requirements Study

3.11 Non-functional Requirements: Privacy and Usability

The interface shall explain what is processed, collect the minimum necessary content, provide configurable retention, and honor eligible deletion requests. It shall use understandable risk labels, keyboard-accessible controls, sufficient contrast, non-color indicators, and calm safety guidance. It shall not promise guaranteed detection or require users to disclose passwords, OTPs, or personal documents in feedback.

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3 System Requirements Study

3.12 Assumptions and Dependencies

CyberShield assumes that users submit material they are authorized to inspect and that external intelligence providers can be delayed, unavailable, or incomplete. OCR quality depends on image resolution, language, compression, and layout. The system depends on authentication, storage, database backup, reputation adapters, OCR capability, and monitoring. Each dependency is isolated so that a known limitation can be reported honestly.

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3 System Requirements Study

3.13 Requirement Prioritization

Core input validation, safe result display, URL analysis, text analysis, screenshot analysis, security controls, and audit records are mandatory for the academic baseline. Export, advanced multilingual NLP, browser-extension integration, organization-specific brand profiles, and SOC workflow integration are future enhancements. Priorities are reviewed with technical feasibility, privacy impact, user benefit, and testing capacity in mind.

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3 System Requirements Study

3.14 Requirement Validation

Requirements are reviewed for clarity, traceability, testability, safety, and consistency with the college project scope. Each requirement maps to at least one unit, integration, security, performance, usability, or acceptance test. Changes are recorded with version identifiers so that an evaluator can determine which scoring policy, model, or interface condition produced a documented result.

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4 System Diagrams

4.1 System Context Diagram

The context model shows that users interact through the browser client, while protected services perform URL, text, and image analysis. External intelligence is accessed by server-side adapters, not directly from the browser.

The diagram is a logical representation intended to support implementation and testing. It identifies responsibility and data movement without locking the project to a specific framework or vendor. Trust boundaries remain between the browser, API, worker processes, external intelligence providers, and protected storage.

Figure: 4.1 System Context Diagram



The implementation is validated with representative benign, ambiguous, and synthetic phishing samples. Acceptance evidence includes correct input validation, a stable result schema, explainable indicators, safe user guidance, and protected audit records. A result is never described as guaranteed safe: where evidence is weak, stale, or unavailable, the interface states that uncertainty and directs the user to verify through an independently known official channel.

4 System Diagrams

4.2 Use Case Diagram

The primary use case is to submit one input modality and receive an evidence-based result. Supporting use cases include account management, feedback, analyst review, and administration.

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Figure: 4.2 Use Case Diagram



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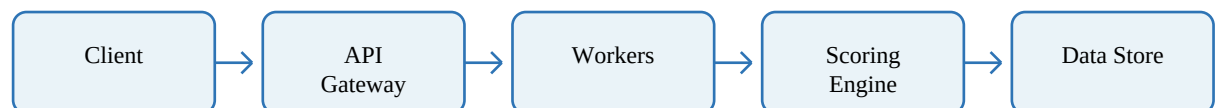
4 System Diagrams

4.3 Level-0 Data Flow Diagram

The Level-0 data flow separates request validation, modality analysis, scoring, and protected storage. It makes clear that raw untrusted inputs are checked before they reach downstream services.

The diagram is a logical representation intended to support implementation and testing. It identifies responsibility and data movement without locking the project to a specific framework or vendor. Trust boundaries remain between the browser, API, worker processes, external intelligence providers, and protected storage.

Figure: 4.3 Level-0 Data Flow Diagram



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4 System Diagrams

4.4 Level-1 Processing Flow

The Level-1 flow describes the common internal pipeline. The same controlled stages apply to URLs, text, and screenshots, while their individual analyzers produce modality-specific evidence.

The diagram is a logical representation intended to support implementation and testing. It identifies responsibility and data movement without locking the project to a specific framework or vendor. Trust boundaries remain between the browser, API, worker processes, external intelligence providers, and protected storage.

Figure: 4.4 Level-1 Processing Flow



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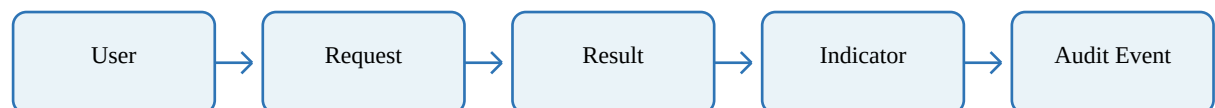
4 System Diagrams

4.5 Entity Relationship Diagram

The entity model connects a user to analysis requests and results, and connects each result to one or more normalized indicators. Audit events remain separate from the result so governance history cannot be overwritten.

The diagram is a logical representation intended to support implementation and testing. It identifies responsibility and data movement without locking the project to a specific framework or vendor. Trust boundaries remain between the browser, API, worker processes, external intelligence providers, and protected storage.

Figure: 4.5 Entity Relationship Diagram



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4 System Diagrams

4.6 Sequence Diagram

The sequence model shows the asynchronous path from browser submission through validation and workers to a scored result. Timeout and failure states are returned as visible limitations rather than hidden technical errors.

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Figure: 4.6 Sequence Diagram



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5 Data Dictionary

5.1 Database Design

CyberShield uses a relational design for identities, requests, results, normalized indicators, feedback, and audit events. Image bytes are not stored directly in normal database rows; protected object storage holds approved assets, while the database retains an opaque reference, integrity hash, consent state, and lifecycle status.

Table 6: Core Data Dictionary

Entity	Selected fields	Purpose
User	user_id, role, consent	Identity and access boundary.
Analysis Request	request_id, modality, status	Tracks one analysis workflow.
Result	score, label, version	Stores an explainable decision.
Indicator	type, severity, source	Stores normalized evidence.
Audit Event	actor, action, time	Governance and investigation trail.

CyberShield treats this capability as a controlled pipeline rather than a single prediction. Input is validated, normalized, and routed only to the services required for the selected modality. Each analyzer returns versioned indicators with severity, evidence source, and confidence. The scoring layer removes duplicate evidence, applies a transparent policy, and records the limitations of unavailable services. This separation makes the behavior easier to test, protects sensitive content, and allows a future model or intelligence-feed change to be reviewed without changing the entire platform.

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5 Data Dictionary

5.2 Core Data Dictionary

The data dictionary defines fields that are necessary to reproduce an analysis safely. Request records store modality, submitted time, status, and a protected input reference. Result records store score, label, confidence, policy version, and limitation. Indicators store type, normalized value or hash, source, severity, and evidence confidence.

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5 Data Dictionary

5.3 Data Retention and Integrity

Retention policy separates transient processing artifacts, user history, audit events, and backups. A content hash supports integrity and duplicate analysis without requiring unrestricted raw content. Deletes are recorded as audit actions; they remove or de-identify eligible content while preserving only the governance information required by policy.

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5 Data Dictionary

5.4 Data Security Controls

Database access uses service accounts with minimum permissions. Sensitive data is encrypted at rest through approved platform controls, and data in transit uses TLS. Queries use parameterized access patterns, exports require authorization, and logs omit credentials, raw session tokens, and discovered secrets.

Table 6: Core Data Dictionary

Entity	Selected fields	Purpose
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6 Result, Discussion and Conclusion

6.1 Functional Results

The planned functional result is an end-to-end workflow that accepts all three input modalities, validates them safely, runs appropriate analyzers, produces an explainable label, and retains an authorized history record. A successful demonstration includes a normal URL, a synthetic phishing SMS/email, and a synthetic screenshot with readable phishing cues.

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6 Result, Discussion and Conclusion

6.2 System Performance Results

Performance evaluation records latency by modality, worker queue time, external-feed timeout behavior, and concurrent request handling. The target is not maximum throughput at any cost; it is predictable response time and a clear limitation when an upstream service is unavailable. Results are interpreted against the selected machine and test workload.

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6 Result, Discussion and Conclusion

6.3 Discussion

CyberShield demonstrates why multi-modal analysis is valuable. A URL checker can find infrastructure cues, but it may miss a persuasive message with no clickable link. Text analysis can recognize social engineering, but it benefits from domain evidence. Screenshot analysis can recover hidden context, but its certainty depends on OCR quality. Combining these sources improves coverage while retaining an explanation of each contribution.

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6 Result, Discussion and Conclusion

6.4 Limitations

No phishing detector can guarantee complete detection. Novel domains may not appear in reputation feeds; legitimate messages can contain urgent language; attackers can use high-quality copied graphics; and OCR can fail on low-resolution or mixed-language images. The platform therefore communicates risk and evidence rather than issuing a legal or forensic verdict. Human verification remains important for high-impact decisions.

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6 Result, Discussion and Conclusion

6.5 Conclusion

CyberShield extends URL analysis into a practical phishing-awareness platform for URLs, SMS/email text, and screenshots. Its contribution is not only the use of cybersecurity, NLP, and image processing, but also the way those methods are connected to safe input handling, transparent scoring, privacy controls, audit records, and user guidance. The documented baseline is suitable for iterative academic implementation and evaluation.

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6 Result, Discussion and Conclusion

6.6 Future Scope

Future work may add multilingual models, email-header analysis, QR-code destination inspection, browser extension support, organization-specific brand profiles, analyst case management, and SOC or ticketing integration. Every enhancement should be assessed for new privacy, security, cost, and false-positive risks before it becomes part of the product.

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